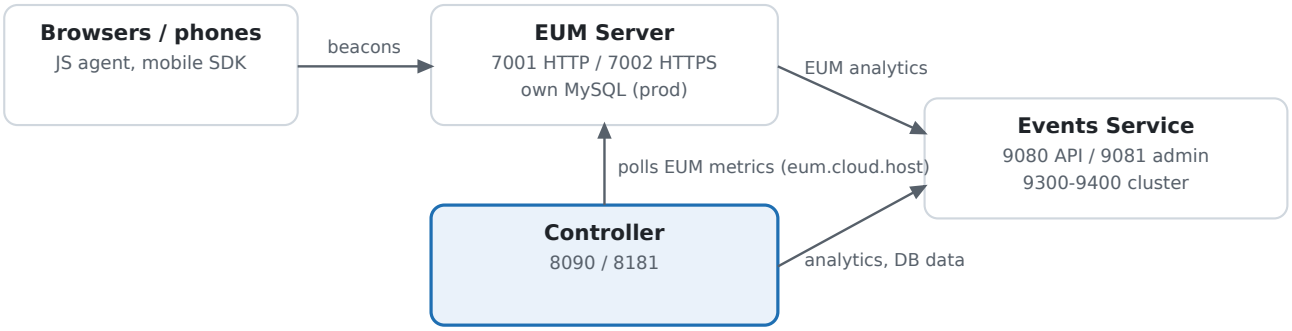
Day 5 — Events Service and EUM Server

Cisco AppDynamics Professional Implementer (500-430 CAPI) · 10-day course · Blueprint 1.4, 2.5, 2.7, 2.8, 3.7

Goal: install, connect, run and upgrade the Events Service and the EUM Server. In the blueprint, "EUEM Processor" (3.7) is the old name for the EUM Server.
Verification: every fact checked against the current Splunk AppDynamics on-prem docs (November 2026).

Block 1 — How the pieces connect



Two different keys connect to the Events Service:

Connection	Setting	Value comes from
Controller → Events Service	appdynamics.on.premise.event.service.key (admin.jsp)	ad.accountmanager.key.controller in events-service-api-store.properties
EUM Server → Events Service	analytics.accountAccessKey (eum.properties)	appdynamics.es.eum.key (shown in the Administration Console)

Block 2 — Install the Events Service (2.5)

Recap: single node or a cluster of **3 or more** dedicated identical machines; heap = half of RAM, **7–31 GB**; SSD; Java 17 bundled.

Profile	Use
Production (default when no profile is given)	4 cores, 12 GB RAM, 128 GB disk minimum
profile=dev	Non-production only (1 core, 1 GB, 2 GB disk)

- **Ports:** 9080 API store · 9081 API store admin · 9300–9400 between cluster nodes.
- **GUI:** EC → Custom Install → Events Service: profile, hosts (1 node or 3+, more can be added later), optional directories (must differ from platform and MySQL data dirs) and ports.

```
bin/platform-admin.sh add-hosts --hosts es01 es02 es03 --credential eskey
bin/platform-admin.sh submit-job --platform-name bank --service events-service \
    --job install --args serviceActionHost=es01      # profile=dev only for a lab
```

Scale up from the embedded Events Service

1. Set up a load balancer.
2. EC: add the hosts.
3. Events Service page → More → **Scale Up Events Service** (don't include the Controller host).

4. On each node: `sudo events-service/latest/bin/tool/tune-system.sh`.
5. Point the Controller at the VIP (settings below). The scale-up job restarts the Controller for you.
6. Old Database Monitoring data stays in the embedded instance unless migrated.

Connect the Controller (admin.jsp → Controller Settings):

- `appdynamics.on.premise.event.service.url` = http://<es-vip>:9080
- `appdynamics.on.premise.event.service.key` = value of `ad.accountmanager.key.controller`

Block 3 — Run the Events Service

Task	How
Start / stop	bin/platform-admin.sh submit-job --platform-name <p> --service events-service --job start | stop
Health (CLI)	bin/platform-admin.sh show-events-service-health — issues + suggested fixes
Health (GUI)	EC → Platforms → <platform> → Events Service: red critical, green normal; Running State shows inactive hosts
Health (HTTP)	https://<es>:9080/_ping → **HTTP 200**
Collect logs	bin/platform-admin.sh retrieve-events-service-logs → events-service.log.zip
Config files	events-service-api-store.properties · events-service.vmoptions (JVM options)

Block 4 — Upgrade the Events Service (2.8)

1. **Upgrade the Enterprise Console first.**
2. No need to stop the Events Service — the EC does it.
3. **GUI:** EC → Events Service page → select hosts → **Upgrade Events Service** → target version (intermediate or latest) → confirm.
4. **CLI:** `bin/platform-admin.sh submit-job --service events-service --job upgrade --platform-name <p>` (log in first with `platform-admin.sh login` if the last session was over a day ago).
5. Failed? Resume with `--args useCheckpoint=true`.
6. Then upgrade the EUM Server, then the Controller.

Block 5 — Install the EUM Server (3.7)

- Installer: `euem-64bit-linux-<ver>.sh` (Windows .exe as Administrator); GUI or silent `-q -varfile response.varfile`. **Not installed by the Enterprise Console.**
- Prerequisites (Linux): libaio. Events Service running if Analytics will be used.

Mode	Rule
Demo	Light testing only; Controller already installed and running **on the same machine**
Production	Own host; installs a **separate MySQL**

- Database passwords (root, eum_user): ASCII only, no ^ / $.
- Keystore password set at install. The installer configures the **HTTP port only** — for HTTPS, a **reverse proxy** in front is recommended.
- Ports: 7001 HTTP · 7002 HTTPS · 3388 EUM MySQL.
- JVM options: Linux EUM/eum-processor/bin/eum-processor · Windows eum-processor-launcher.vmoptions.

Block 6 — Connect, license and verify the EUM Server

1. Provision the license (single-tenant; MySQL running; must run from eum-processor):

```
cd $EUM_HOME/eum-processor
./bin/provision-license <PATH_TO_EUM_LICENSE>     # one license at a time
```

Multi-tenant: provision the same way, then admin.jsp → **Account Settings** → edit each account → **EUM License Key** and **EUM Account Name**.

2. Connect to the Events Service — eum-processor/bin/eum.properties, then restart:

```
analytics.enabled=true
analytics.serverScheme=http
analytics.serverHost=<es-vip>
analytics.port=9080
analytics.accountAccessKey=<value of appdynamics.es.eum.key>
```

3. Point the Controller at the EUM Server (admin.jsp → Controller Settings):

Property	Value	Purpose
eum.cloud.host	http://<eum>:7001	Controller **polls** EUM metrics
eum.beacon.host	http://<eum>:7001	JS agent beacons over HTTP
eum.beacon.https.host	https://<eum>:7002	Beacons over HTTPS
eum.mobile.screenshot.host	http://<eum>:7001	Mobile screenshots

4. Start, stop, verify (Linux):

```
cd <EUM>/eum-processor && bin/eum.sh start          # bin/eum.sh stop
curl http://<eum>:7001/eumaggregator/ping           # replies "ping"
# /eumcollector/ping also works
```

- EUM MySQL is started/stopped with orcha-manager playbooks (start-mysql.orcha / stop-mysql.orcha). Windows: both are services.
- **NTP** on both the EUM Server and the Controller.

Block 7 — Upgrade the EUM Server (2.7)

1. **Stop the EUM Server — keep its MySQL running.**
2. Back up: whole EUM directory; <EUM_HOME>/jre/lib/security/cacerts (outside the install path); eum_db (mysqldump --add-drop-database); custom eum.properties settings.
3. Set a password in eum-processor/bin/liquibase.properties; copy orcha.properties to the new installation.
4. Run the new EUM installer.
5. After: re-apply custom eum.properties (incl. analytics.accountAccessKey); for HTTPS without a proxy set processorServer.httpsProduction=true and processorServer.httpsPort; restore cacerts; copy back bin/credential.scs.
6. Restart and verify: `curl http://<eum>:<port>/eumcollector/ping` → ping.

   Platform order: EC → Events Service → **EUM Server** → Controller.

Paper lab (~45 min)

1. **Events Service cluster:** steps and commands for a production 3-node cluster on es01–es03 behind VIP es-vip.bank.local, and connecting the Controller.
2. **Production EUM Server** on eum01.bank.local: install choices, license step, eum.properties lines and the four Controller properties.

3. **Troubleshooting:** Browser RUM is configured but no browser data appears in the Controller — your checklist, in order.
4. **EUM upgrade:** step-by-step plan including backups and verification.

Day 5 Quiz — with answer key

Cover the answer lines and test yourself first.

Q1. Which value goes into the Controller's appdynamics.on.premise.event.service.key?

A. appdynamics.es.eum.key
B. ad.accountmanager.key.controller from events-service-api-store.properties
C. Controller license key
D. EUM account name
Answer: B — Controller → ES uses ad.accountmanager.key.controller; EUM uses appdynamics.es.eum.key.

Q2. No profile specified when installing the Events Service — what is installed?

A. Dev
B. Production
C. Embedded on the Controller host
D. The install fails
Answer: B — Default = production (4 cores, 12 GB, 128 GB).

Q3. Which port range do cluster nodes use to talk to each other?

A. 9080–9081
B. 9300–9400
C. 7001–7002
D. 3377–3388
Answer: B — 9080 API, 9081 admin, 9300–9400 between nodes.

Q4. Before upgrading an Events Service cluster through the EC, you must:

A. Stop every node manually
B. Upgrade the EC first (the EC stops the ES for you)
C. Upgrade the Controller first
D. Delete old indices
Answer: B — Upgrade the EC first; no manual stop needed.

Q5. An Events Service upgrade job failed halfway. Resume how?

A. Reinstall
B. Re-run with --args useCheckpoint=true
C. Run tune-system.sh
D. Roll back the Controller
Answer: B — useCheckpoint=true resumes the job.

Q6. Where must ./bin/provision-license run on a Linux EUM Server?

A. Any directory
B. $EUM_HOME/eum-processor
C. <controller_home>/bin
D. $EUM_HOME/orcha
Answer: B — From eum-processor, with MySQL running, one license at a time.

Q7. Which property tells the Controller where to poll EUM metrics?

A. eum.beacon.host
B. eum.cloud.host
C. eum.mobile.screenshot.host
D. appdynamics.on.premise.event.service.url
Answer: B — eum.cloud.host = polling; beacon hosts = where browsers send beacons.

Q8. Multi-tenant Controller: after provisioning EUM licenses, what else?

A. Nothing
B. Enter EUM License Key + EUM Account Name per account in Account Settings
C. Restart the ES
D. Reinstall EUM in demo mode
Answer: B — Edit each account in admin.jsp → Account Settings.

Q9. Upgrading a production EUM Server — how do you handle its MySQL at the start?

A. Stop EUM and MySQL
B. Stop EUM, keep MySQL running
C. Drop eum_db
D. Move MySQL to the Controller
Answer: B — MySQL must stay up to dump eum_db and for the upgrade.

Q10. Which URL confirms an EUM Server is running?

A. /controller/rest/serverstatus on 7001
B. /eumaggregator/ping (or /eumcollector/ping) → 'ping'
C. https://<eum>:9080/_ping
D. http://<eum>:9191
Answer: B — 9080/_ping is the Events Service; serverstatus is the Controller; 9191 is the EC.

Quick-review cheat sheet

Remember	Value
ES ports	9080 API · 9081 admin · 9300–9400 cluster
ES nodes / heap	1 or ≥ 3 · heap ½ RAM, 7–31 GB · default profile = production
Controller → ES	event.service.url = http://<vip>:9080 · key = ad.accountmanager.key.controller
EUM → ES	eum.properties analytics.* · accountAccessKey = appdynamics.es.eum.key
ES health	show-events-service-health · https://<es>:9080/_ping → 200
ES upgrade	EC first · no manual stop · --job upgrade · resume useCheckpoint=true
EUM install	Own installer (not EC) · Demo = same host as Controller · Production = own host + own MySQL
EUM ports	7001 HTTP · 7002 HTTPS · 3388 MySQL
EUM license	cd eum-processor · ./bin/provision-license <file> · multi-tenant → Account Settings key + name
Controller eum.* props	cloud.host (poll) · beacon.host · beacon.https.host · mobile.screenshot.host
EUM verify	/eumaggregator/ping or /eumcollector/ping → ping
EUM upgrade	Stop EUM, keep MySQL · back up dir, cacerts, eum_db · new installer · restore config + credential.scs
Order	EC → Events Service → EUM Server → Controller

Sources

- Install the Events Service on Linux
- Administer the Events Service
- Upgrade the Multi-Node Events Service Cluster for Linux
- Events Service General Requirements
- Install a Production EUM Server
- Install a Demo EUM Server
- Provision EUM Licenses

- Start and Stop the EUM Server and Database
- Upgrade the Production EUM Server
- Upgrade Platform Components

Paper lab — model answers

How an implementer would answer each Day 5 paper-lab task, with the rule behind each step.

1. Production 3-node Events Service behind es-vip.bank.local

Prepare: three dedicated, identical Linux hosts (same directories, user, hardware), SSD ≥ 1 TB, clocks synced. With 64 GB RAM per node the heap is **31 GB** (cap). Open 9080/9081 to the EC and Controller, and 9300–9400 between nodes.

Install:

```
bin/platform-admin.sh add-credential --credential-name eskey --type ssh \
    --user-name appduser --ssh-key-file /home/appduser/.ssh/appd.pem
bin/platform-admin.sh add-hosts --hosts es01 es02 es03 --credential eskey
# EC GUI → Custom Install → Events Service: production profile (no profile=dev),
#   hosts es01, es02, es03  → Install, watch the Jobs page
```

- **Load balancer:** VIP es-vip.bank.local:9080 → es01–es03:9080; health check https://<node>:9080/_ping expecting 200.
- **Connect the Controller** (admin.jsp → Controller Settings): appdynamics.on.premise.event.service.url = http://es-vip.bank.local:9080; appdynamics.on.premise.event.service.key = ad.accountmanager.key.controller from events-service-api-store.properties.
- **Verify:** platform-admin.sh show-events-service-health; EC Events Service page all green; allow a few minutes for account information to sync.

2. Production EUM Server on eum01.bank.local

1. Install libaio; chmod 775 euem-64bit-linux-<ver>.sh and run it (or -q -varfile response.varfile).
2. Choose **Production** mode (separate host → separate MySQL). Set root and eum_user passwords (ASCII, no ^ / $) and the keystore password.
3. License: cd $EUM_HOME/eum-processor; check MySQL is running (ps aux | grep mysql); ./bin/provision-license /opt/licenses/license.lic.
4. eum.properties: analytics.enabled=true · analytics.serverScheme=http · analytics.serverHost=es-vip.bank.local · analytics.port=9080 · analytics.accountAccessKey=<appdynamics.es.eum.key>. Then bin/eum.sh stop and bin/eum.sh start.
5. Controller (admin.jsp → Controller Settings): eum.cloud.host = http://eum01.bank.local:7001 · eum.beacon.host = http://eum01.bank.local:7001 · eum.beacon.https.host = https://eum01.bank.local:7002 · eum.mobile.screenshot.host = http://eum01.bank.local:7001.
6. NTP on EUM and Controller. Internet browsers send HTTPS beacons → put a reverse proxy with a CA-signed certificate in front.
7. Verify: curl http://eum01.bank.local:7001/eumaggregator/ping → ping.

3. Browser RUM configured but no data in the Controller — checklist

#	Check	Why
1	EUM Server up: /eumaggregator/ping returns ping	No server, no data
2	EUM license provisioned (MySQL running, run from eum-processor); multi-tenant → key + account name in Account Settings	Unlicensed EUM accepts no data
3	JS snippet uses the right app key and its beacon URLs point to the on-prem EUM Server (HTTPS pages need the HTTPS beacon URL)	Beacons may be going to the SaaS cloud or blocked as mixed content
4	Browsers can reach the beacon host: DNS, firewall, reverse proxy, certificate	Beacons come from users' browsers, not from your servers
5	Controller eum.cloud.host points to the EUM Server	The Controller polls EUM for metrics
6	NTP in sync on EUM Server and Controller	Clock skew breaks data alignment
7	For Browser Analytics: eum.properties analytics.* settings and key; Events Service healthy (_ping 200)	Analytics data goes EUM → Events Service

4. EUM Server upgrade plan

Step	Action
0. Order	Do it after the EC and Events Service upgrades, before the Controller
1. Stop	Stop the EUM Server (bin/eum.sh stop) — **keep its MySQL running**
2. Back up	Whole EUM directory · jre/lib/security/cacerts (outside the install path) · mysqldump eum_db --add-drop-database · note custom eum.properties
3. Prepare	Set a password in eum-processor/bin/liquibase.properties · copy orcha.properties to the new installation
4. Install	Run the new euem-64bit-linux installer
5. Restore	Re-apply custom eum.properties (incl. analytics.accountAccessKey) · HTTPS without proxy → processorServer.httpsProduction=true + httpsPort · restore cacerts · copy bin/credential.scs
6. Verify	Restart; curl http://eum01.bank.local:7001/eumcollector/ping → ping; check browser data and analytics in the Controller
Rollback	Restore the EUM directory backup and the eum_db dump